

MPhil Thesis: Empirical Results

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1 RQ1: Overall Innovation Response

1.1 Patents

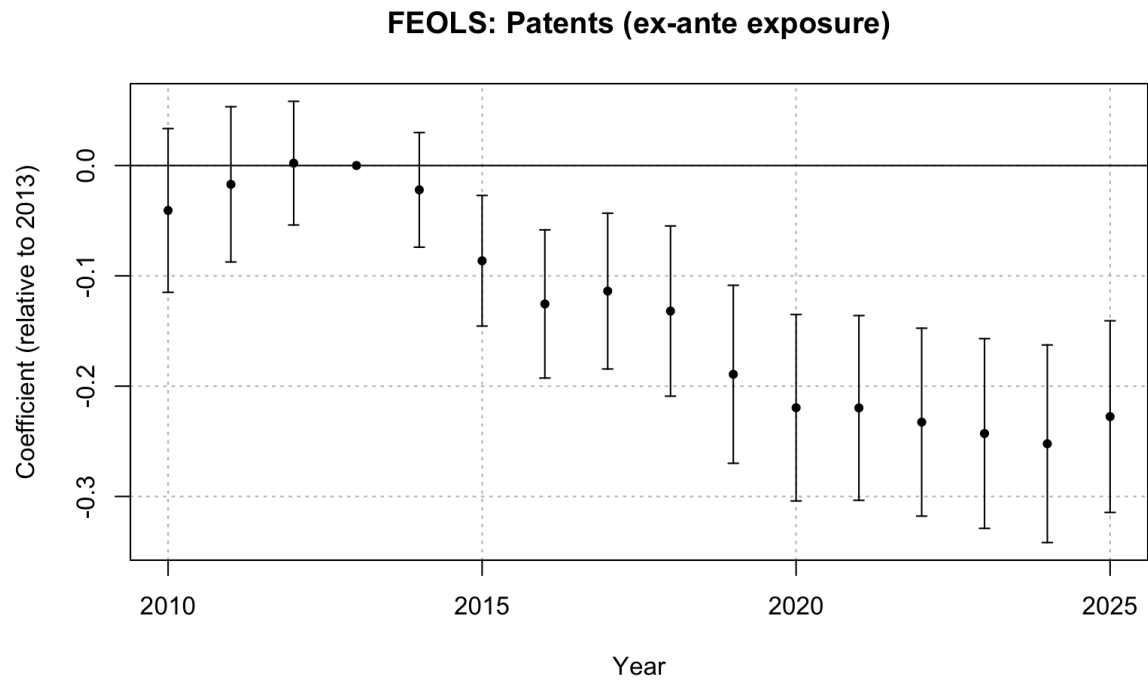


Figure 1: FEOLS event study: patents, ex-ante Medicaid exposure

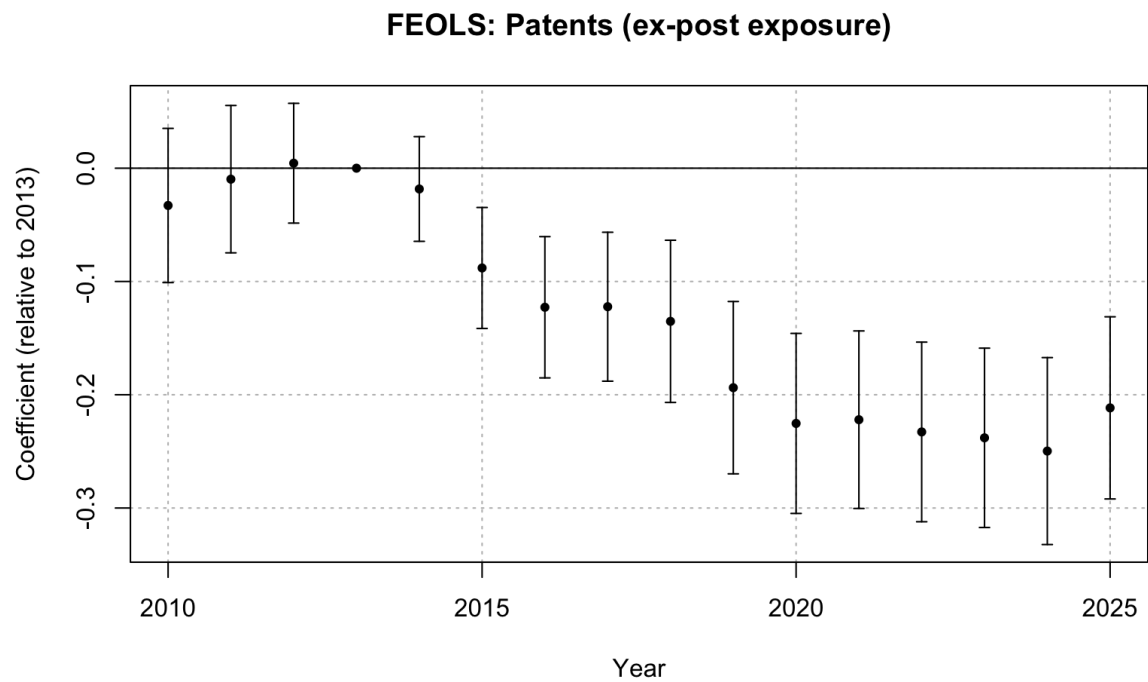


Figure 2: FEOLS event study: patents, ex-post Medicaid exposure

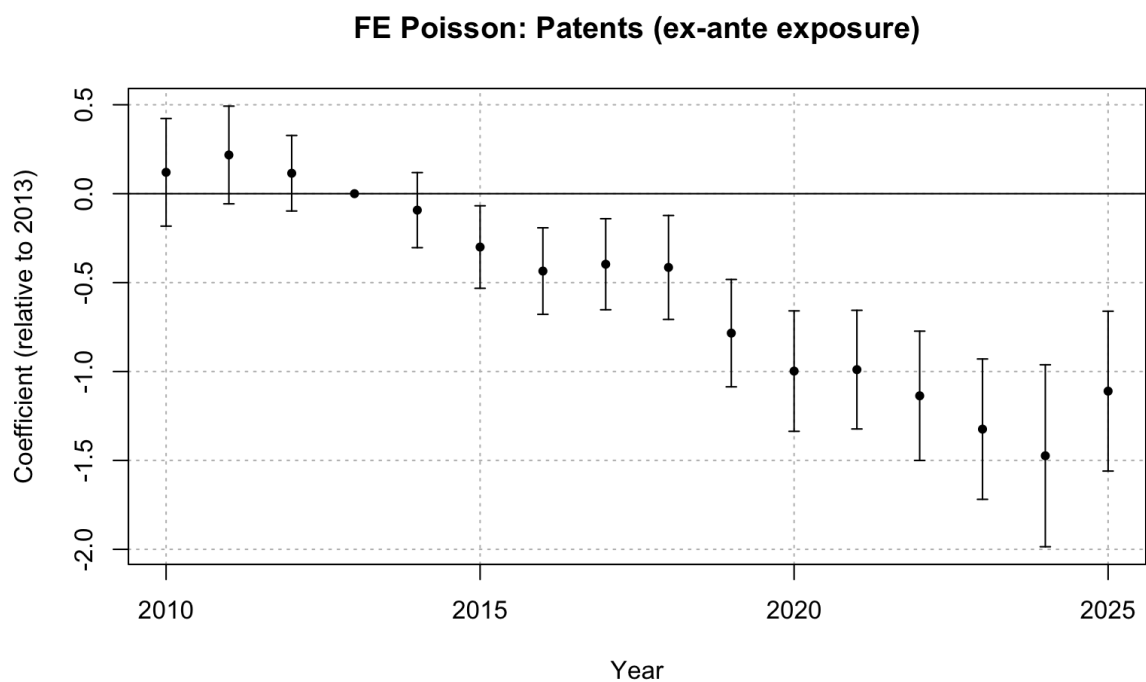


Figure 3: FE Poisson event study: patents, ex-ante Medicaid exposure

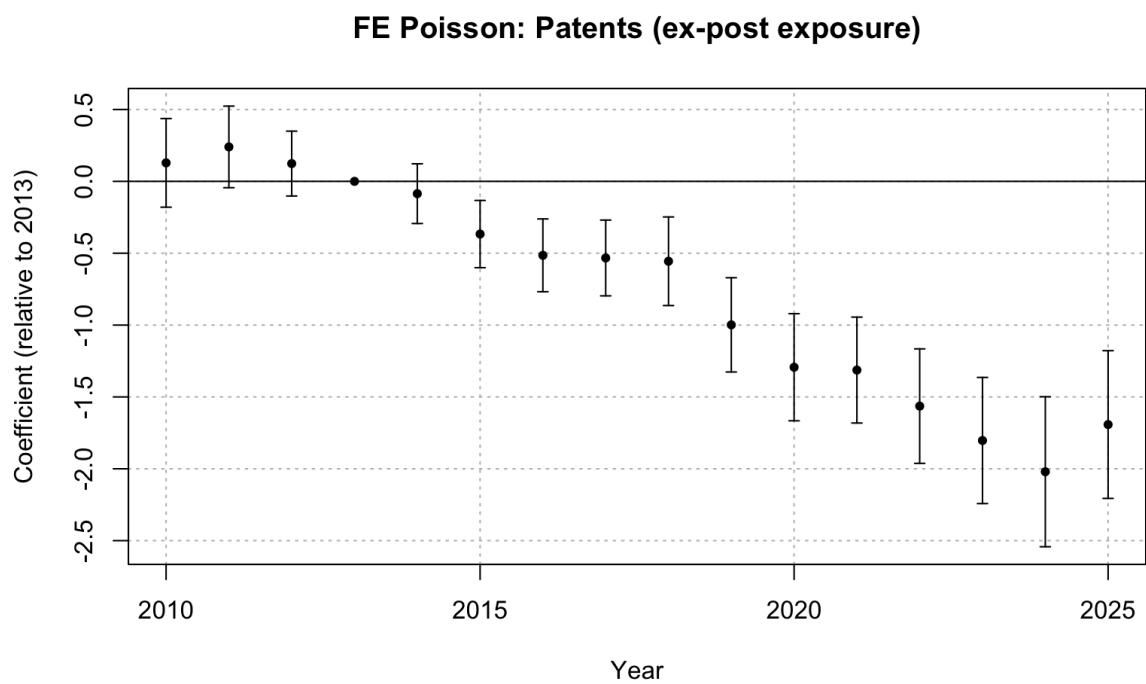


Figure 4: FE Poisson event study: patents, ex-post Medicaid exposure

The pre-expansion coefficients between 2010-2012 are small and statistically insignificant across all four specifications, which is consistent with the assumption of parallel trends in firm patenting behaviour between areas with differential Medicaid market shares. Under the FEOLS specification with ex-ante exposure, the post-expansion coefficient is insignificant in 2014 and then becomes significant in 2015 at -0.086 and then begins to grow more significantly negative over time with a peak of -0.252 in 2024. Since the exposure to Medicaid is measured as a therapeutic classes market share within Medicaid, the estimated coefficients corresponds to a one-hundred percentage point increase in Medicaid share. Consequently, a therapeutic class with a 10 percentage-point higher Medicaid market share would be associated with 0.025 fewer patents per firm-class-year by 2024, this corresponds to about 16 percent of the sample mean patent count (0.16). Noticeably, the coefficients stabilise between 2020 and 2025 in the range of -0.220 to -0.252 , indicating that the negative response persists as opposed to reverting.

The gradual onset of the effect is expected given that patent outcomes are recorded by the date of award rather than the application date. Thus, patents observed shortly after 2014 may include applications prior to the ACA, so the timing should not be interpreted as the exact timing of underlying R&D decisions. Additionally, the effect also reflects transition dynamics when making R&D decisions. Firms would not immediately adjust their innovation decisions in response to policy shifts, and pricing in the new Medicaid market conditions into R&D plans may take time given the long development horizons involved in pharmaceutical innovation. Both of these factors therefore contribute towards the increasing magnitude of the coefficients where stabilisation is seen between 2020 and 2025.

The FE Poisson estimates are consistent with this result where the negative trend is observed post-ACA with the estimate peaking at -1.474 by 2024 under ex-ante exposure and then stabilising by 2025. Specifically, in 2024 a ten percentage point increase in Medicaid exposure corresponds to approximately a 14.74 percent decrease in expected patents. The ex-ante and ex-post specifications produce similar results throughout, suggesting that the qualitative findings are not driven by the construction of the exposure measure.

1.2 FDA Approvals

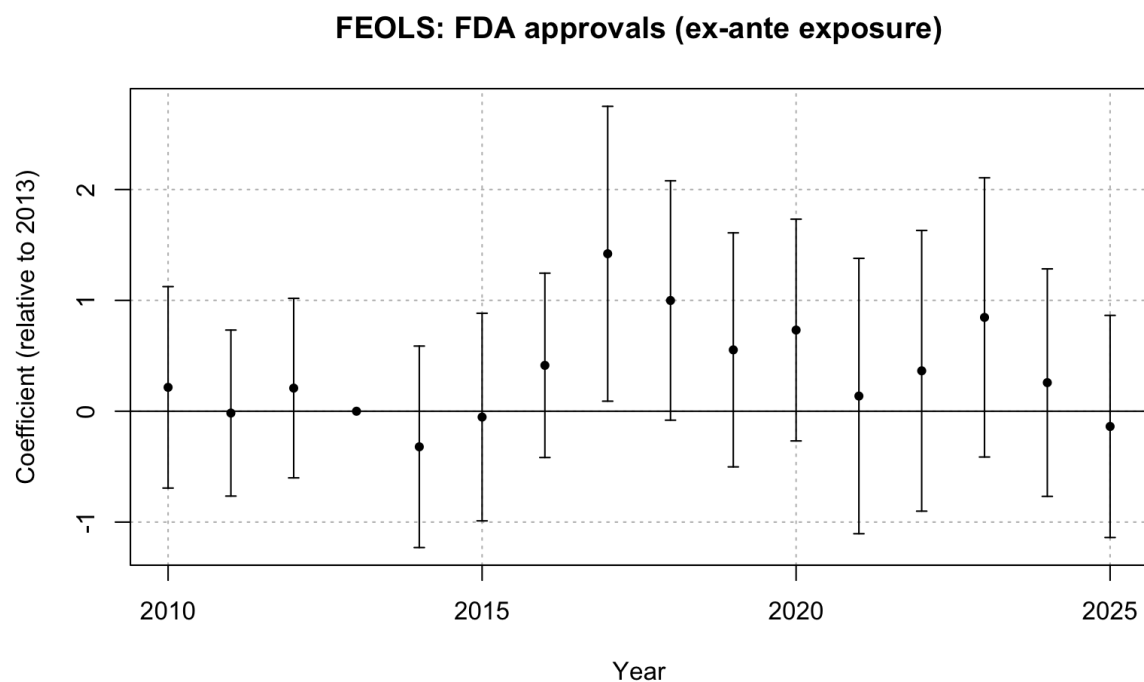


Figure 5: FEOLS event study: FDA approvals, ex-ante Medicaid exposure

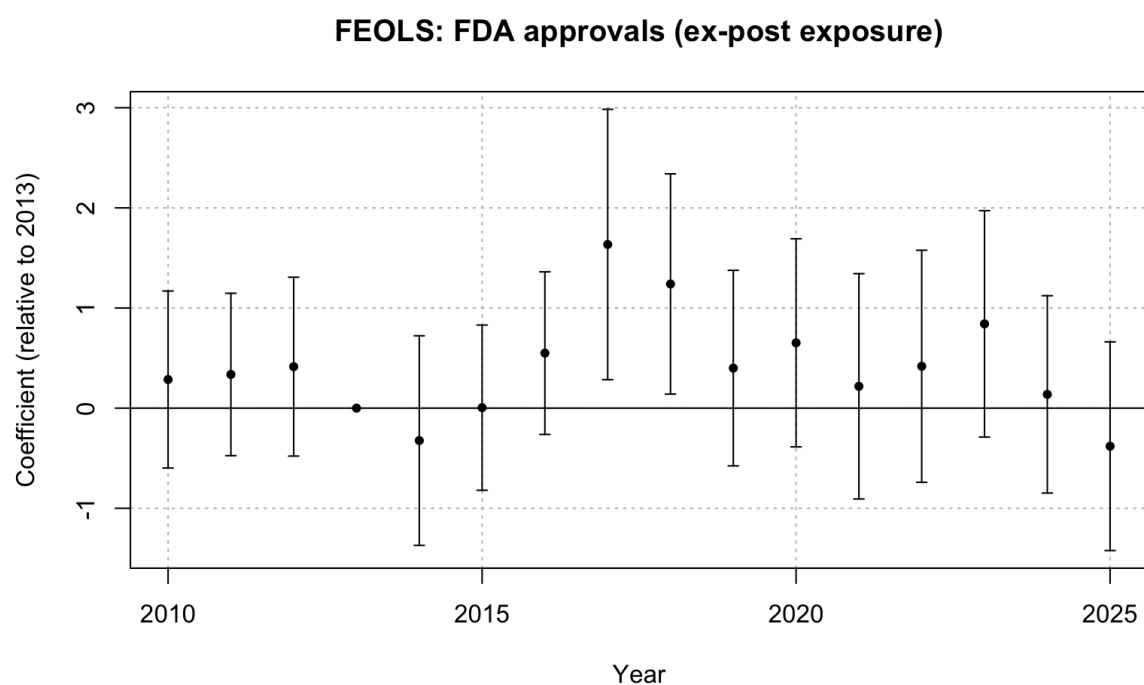


Figure 6: FEOLS event study: FDA approvals, ex-post Medicaid exposure

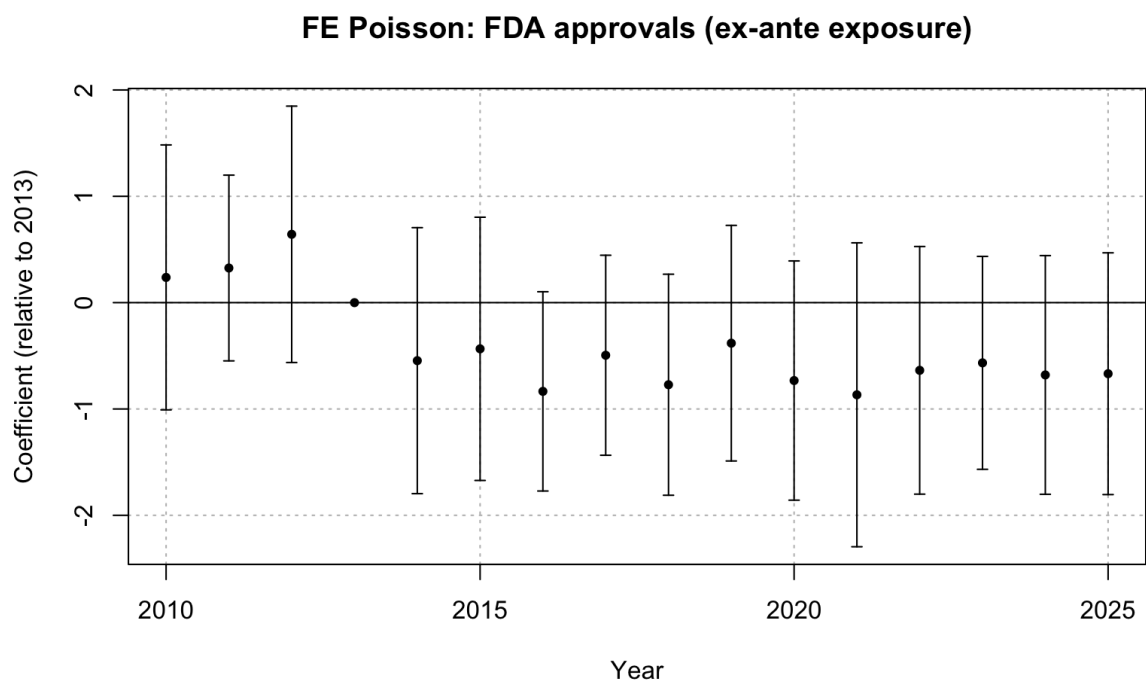


Figure 7: FE Poisson event study: FDA approvals, ex-ante Medicaid exposure

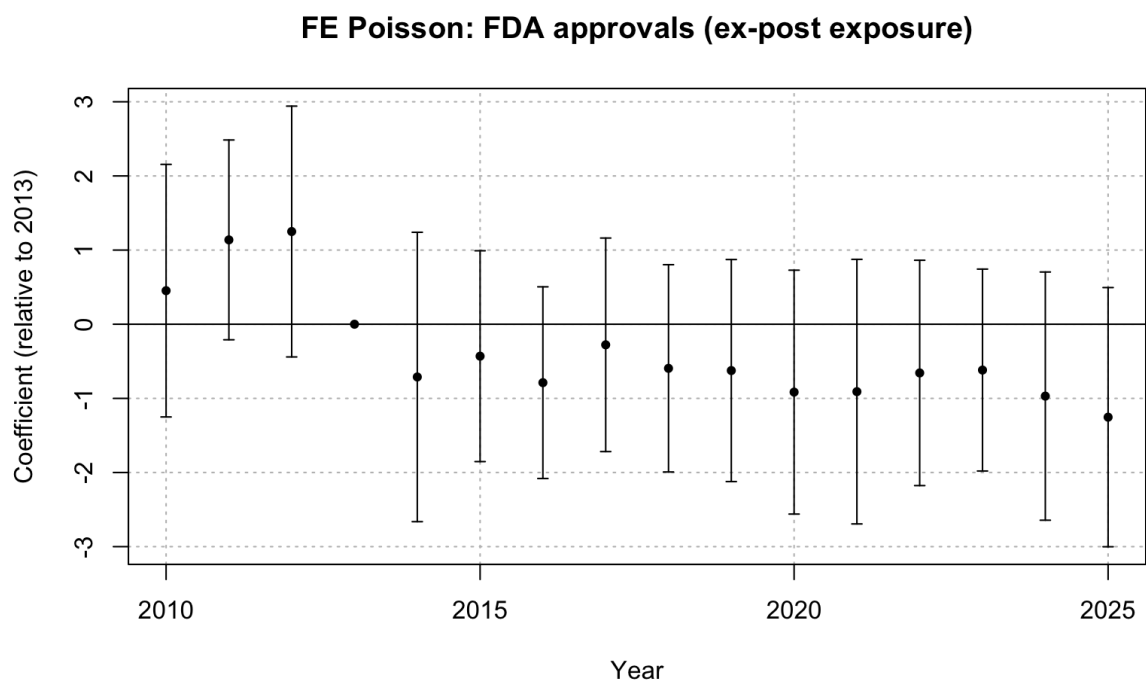


Figure 8: FE Poisson event study: FDA approvals, ex-post Medicaid exposure

The results from FDA approvals contrast with the patent findings. Across all specifications, the post-expansion coefficients are statistically insignificant with wide confidence intervals. Whereas the patent FEOLS estimates grow monotonically from -0.086 in 2015 to -0.252 in 2024, the FDA FEOLS estimates fluctuate between -0.321 and 1.421 without a sustained direction, with isolated significance in 2017 and 2018 that is consistent with sampling variation given the number of post-expansion periods tested. The FE Poisson estimates are uniformly negative but individually insignificant. The two specifications therefore point in opposite directions while agreeing on insignificance, indicating that the ACA expansion had inconclusive effects on FDA approvals.

The difference between the patent and FDA results is consistent with the innovation response emerging earlier in the R&D pipeline. Pharmaceutical development spans over a decade from molecule discovery to drug approval (SOURCE), and patenting activity captures an earlier stage of this process than regulatory approval. Therefore, the ACA policy shock may plausibly affect patenting without yet generating a response in FDA approvals. Changes in patenting decisions would feed through into drugs under active development, subsequently appear in clinical trial activity, and once the affected drug cohort reaches the end of the regulatory process would it feed into FDA approvals. Under this interpretation, the insignificant response of FDA approvals reflects that insufficient time has passed for the upstream adjustment to result in detectable changes at the approval stage as opposed to contradicting the patent findings.

This is consistent with Garthwaite et al. (2021), who study the response of clinical trial activity two-year post Medicaid expansion and find an insignificant innovation response. Clinical trial activity generally occurs later in the development process than early patenting activity, and given that the patent event-study displays a gradual build up over nearly a decade, a two-year window is likely too short to detect a response even at the clinical trial stage.

2 RQ2: Medicaid Demand and PDL Exposure

2.1 Patents

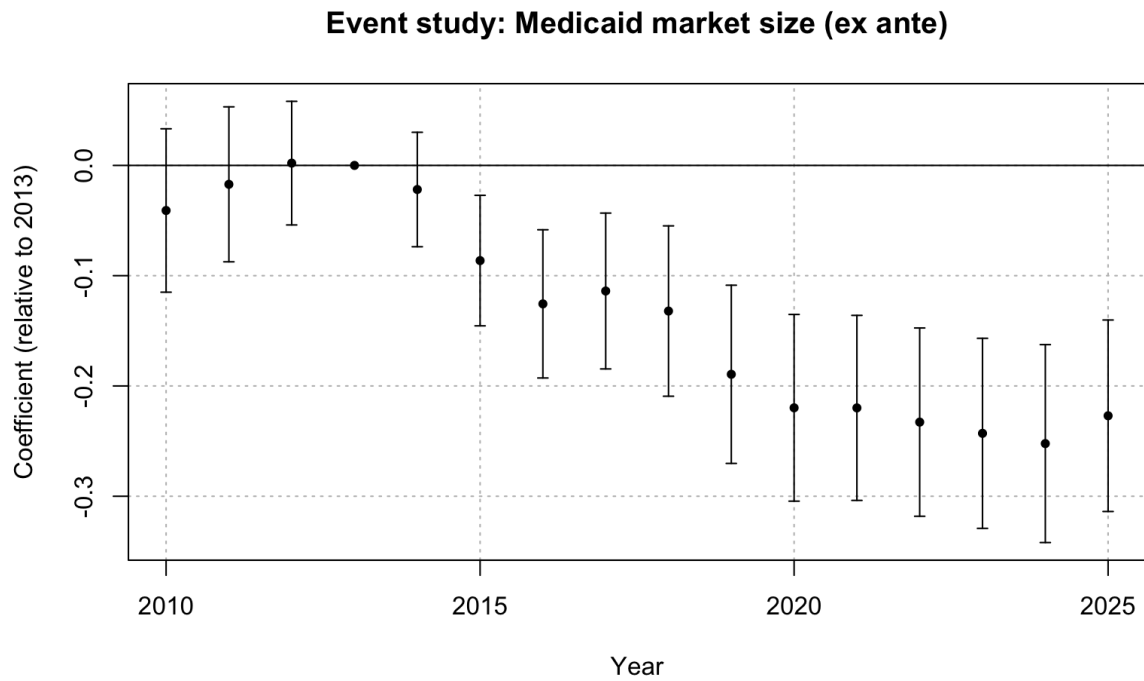


Figure 9: FEOLS event study: patents, Medicaid exposure (ex-ante)

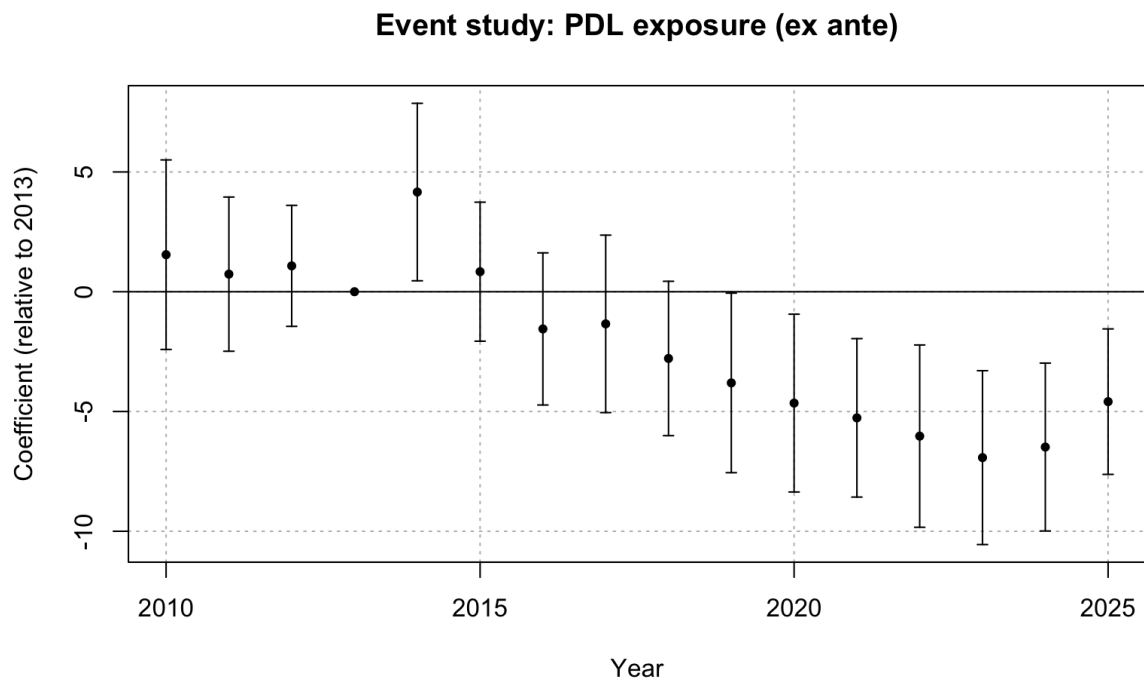


Figure 10: FEOLS event study: patents, PDL exposure (ex-ante)

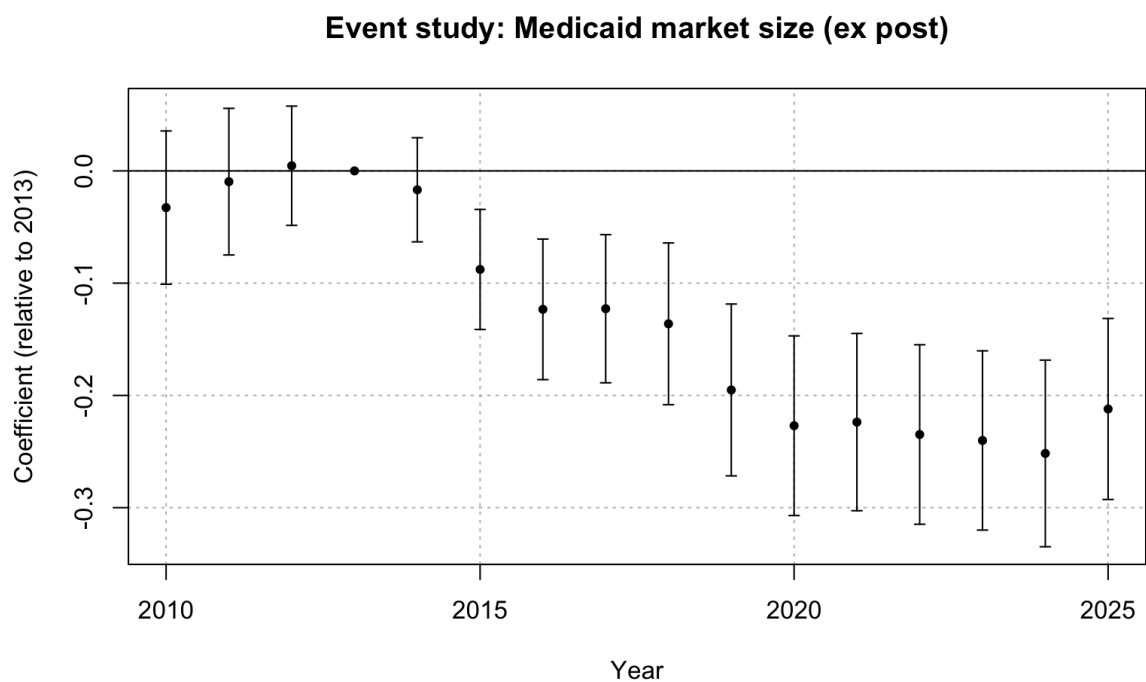


Figure 11: FEOLS event study: patents, Medicaid exposure (ex-post)

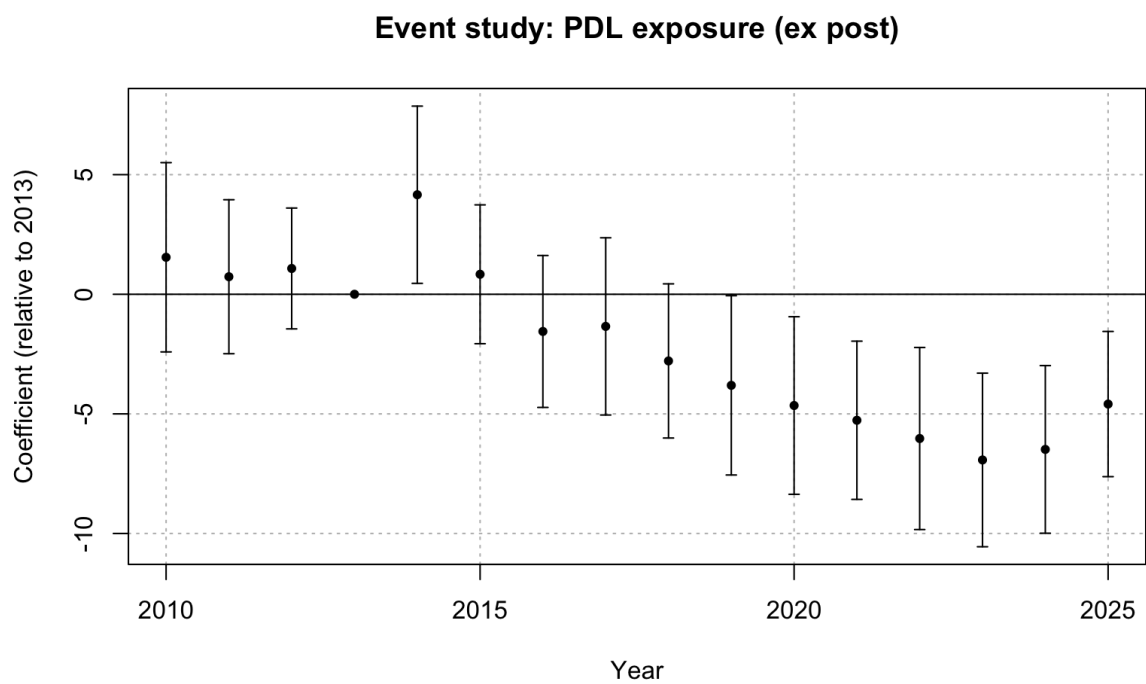


Figure 12: FEOLS event study: patents, PDL exposure (ex-post)

The Medicaid exposure coefficients in the joint specification are consistent with the findings of the previous research question, with insignificant pre-period estimates and negative post-expansion coefficients growing from -0.086 in 2015 to -0.252 by 2024. Controlling for firm exposure to the PDL formulary yields larger Medicaid exposure coefficient estimates. This suggests that the RQ1 Medicaid coefficient partially absorbed variation associated with procurement exposure when PDL exposure was omitted. Including both measures therefore provides a cleaner comparison between the Medicaid demand response and the additional response associated with firms' exposure to PDL institutions. Since PDL exposure is a proxy for procurement exposure rather than a direct measure of rebate extraction, the estimates should not be interpreted as a full decomposition of the demand and pricing channels.

The PDL exposure coefficients point towards an additional source for the decline in post-expansion patenting. The pre-expansion PDL coefficients are positive but insignificant, providing support for the parallel trends assumption. The isolated positive estimate in 2014 is not persistent and should not be interpreted as a full effect of the ACA. From 2016 onwards, the PDL coefficients turn negative and become significant from 2019 onwards and grow to -6.925 by 2023. This indicates that firms with greater formulary exposure experienced an additional decline in patenting relative to the Medicaid demand effect. The ex-ante and ex-post specifications produce nearly identical results throughout.

This result should not be interpreted as showing that preferred placement itself reduces innovation incentives. The PDL exposure measure instead depicts the pre-expansion share of a firm's portfolio already on the PDL prior to Medicaid expansion, thus the negative coefficient indicates that firms more deeply embedded in Medicaid's formulary system before the reform experienced a stronger negative innovation response once Medicaid demand increased. This is consistent with these firms being more exposed to the procurement pressures that intensified following the expansion, even though securing preferred placement remains beneficial relative to failing one of the PDL stages.

2.2 FDA Approvals

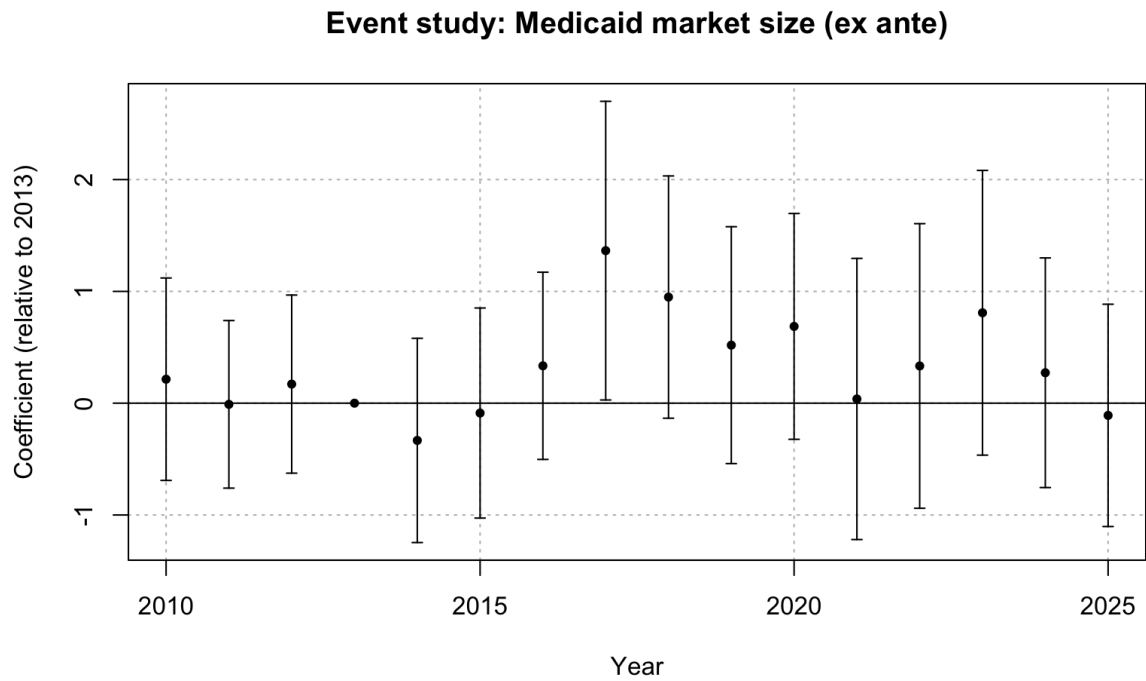


Figure 13: FEOLS event study: FDA approvals, Medicaid exposure (ex-ante)

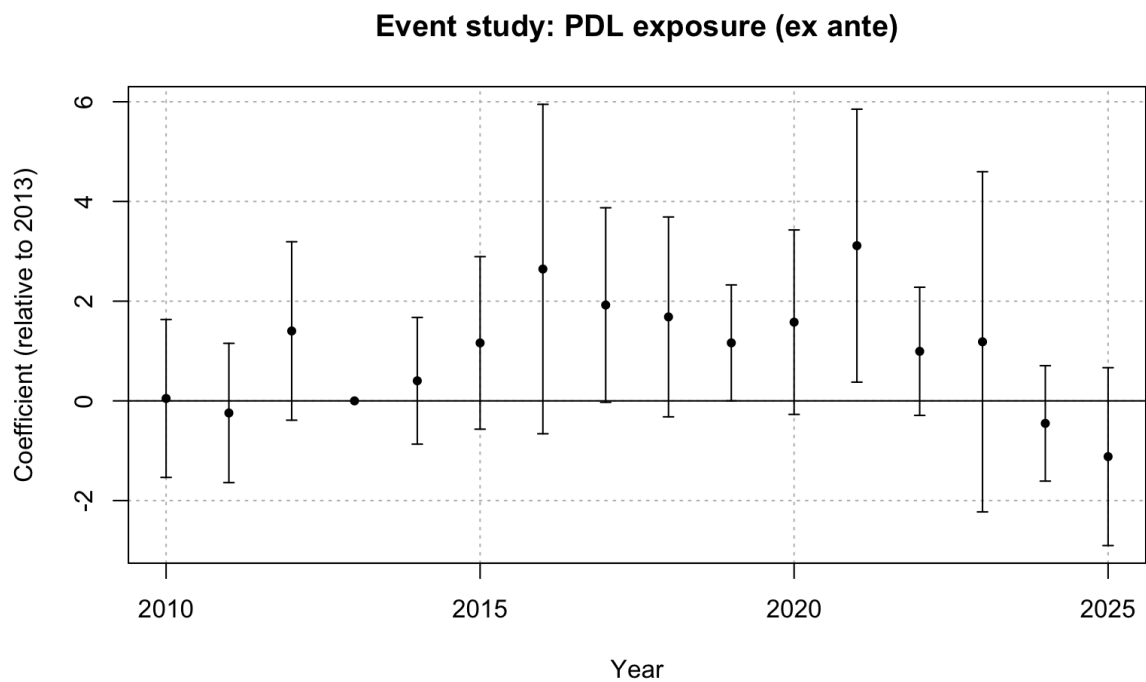


Figure 14: FEOLS event study: FDA approvals, PDL exposure (ex-ante)

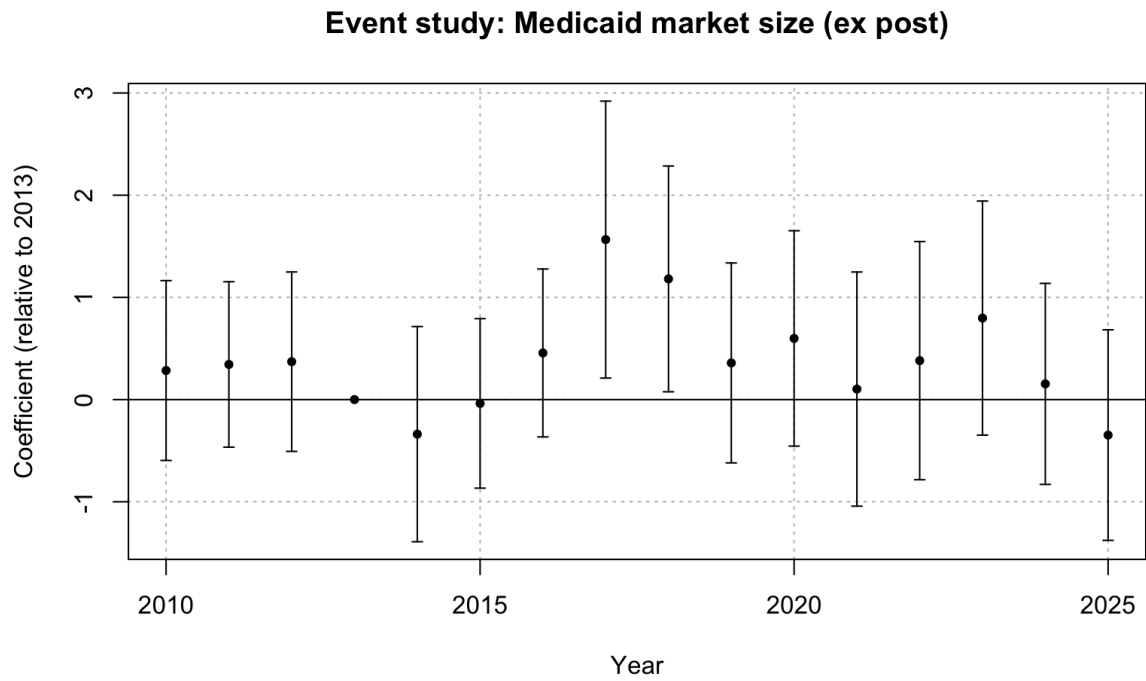


Figure 15: FEOLS event study: FDA approvals, Medicaid exposure (ex-post)

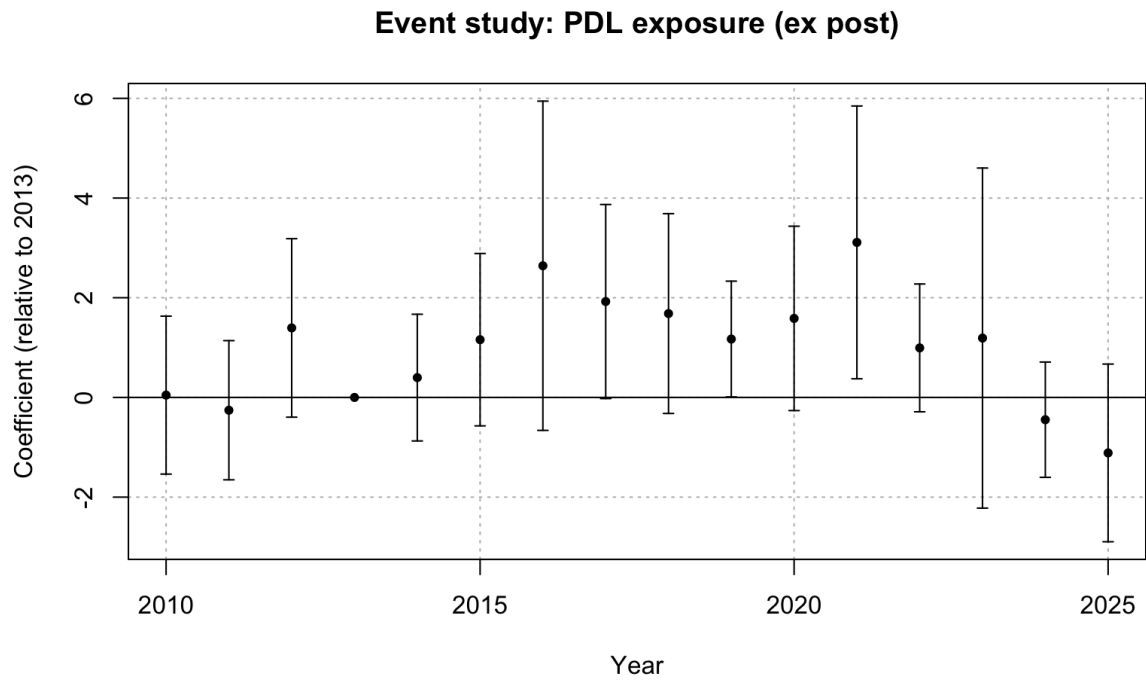


Figure 16: FEOLS event study: FDA approvals, PDL exposure (ex-post)

Both the Medicaid exposure and PDL exposure coefficients for FDA approvals are statistically insignificant throughout the post-expansion period. There are isolated exceptions for this, with 2017 and 2018 for the Medicaid coefficients and 2019 and 2021 for the PDL coefficients, but these do not form a sustained pattern. Although a significant significant PDL effect is recorded for patents, the absence of a corresponding effect on FDA approvals is consistent with the broader interpretation that the innovation response first appears in upstream R&D activity and has not yet propagated to the final stage of the R&D pipeline.

3 RQ3: Therapeutic-Class Heterogeneity

3.1 Patents

	FEOLS: Ex Ante	FEOLS: Ex Post
Medicaid Share $\times$ ATC = A	−0.165*** (0.033)	−0.157*** (0.031)
Medicaid Share $\times$ ATC = B	0.134*** (0.024)	0.151*** (0.027)
Medicaid Share $\times$ ATC = C	−0.123*** (0.019)	−0.094*** (0.015)
Medicaid Share $\times$ ATC = D	−0.030 (0.108)	−0.017 (0.062)
Medicaid Share $\times$ ATC = G	0.037 (0.056)	0.032 (0.049)
Medicaid Share $\times$ ATC = H	0.876*** (0.136)	0.619*** (0.096)
Medicaid Share $\times$ ATC = J	0.055 (0.054)	0.035 (0.034)
Medicaid Share $\times$ ATC = L	0.346 (0.452)	0.443 (0.578)
Medicaid Share $\times$ ATC = M	−0.336*** (0.076)	−0.212*** (0.048)
Medicaid Share $\times$ ATC = N	−0.125*** (0.028)	−0.174*** (0.039)
Medicaid Share $\times$ ATC = R	0.005 (0.003)	0.005 (0.003)
Medicaid Share $\times$ ATC = S	0.093*** (0.017)	0.105*** (0.020)
Num. obs.	19 617 728	19 617 728
R2	0.247	0.247

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The triple-difference estimates reveal substantial heterogeneity across therapeutic areas. Significant negative responses are observed in patenting behaviour for alimentary tract and metabolism, cardiovascular, musculo-skeletal, and nervous system drugs. Since RQ1 identifies an aggregate decline in patenting following the Medicaid expansion, these classes appear to drive that overall result. These are among the highest-volume therapeutic areas within Medicaid and cover widely prescribed drug categories such as diabetes and gastrointestinal medications, anti-inflammatories and antidepressants. As shown later in the theoretical model, and in the NDC counts in Table 1, these classes typically have greater therapeutic substitutability, consistent with stronger formulary competition and rebate pressure. Consequently, this combination of high prescription volumes and stronger substitutability is consistent with the procurement channel driving deeper innovation reductions in these areas.

Positive patenting responses are observed for blood and blood-forming organs, hormonal

preparations, and sensory organs. This indicates that the standard market size prediction holds for these therapeutic classes suggesting the demand effect outweighs the disincentives associated with Medicaid's procurement channel. Insignificant coefficients are observed for the remaining therapeutic classes suggesting the demand and procurement effects offset one another, or the expansion may be unimportant for patenting decisions in these areas. The ex-post specification yields comparable results

The presence of positive class-level estimates indicates that some therapeutic areas experience a demand-driven innovation response, suggesting that the standard market-size effect is not absent in the Medicaid setting. Instead, the aggregate negative patenting response identified in RQ1 is concentrated in therapeutic classes where high prescription volumes and therapeutic substitutability strengthen the procurement channel. Therefore, the overall decline in patenting reflects the two opposing forces operating differentially across therapeutic areas, rather than a uniform reduction in pharmaceutical innovation.

3.2 FDA Approvals

	FEOLS: Ex Ante	FEOLS: Ex Post
Medicaid Share $\times$ ATC = A	1.211** (0.391)	1.301** (0.420)
Medicaid Share $\times$ ATC = B	-0.396 (2.567)	-0.121 (0.784)
Medicaid Share $\times$ ATC = C	0.229 (0.171)	0.373 (0.278)
Medicaid Share $\times$ ATC = D	0.707** (0.270)	0.881** (0.337)
Medicaid Share $\times$ ATC = G	-2.516 (2.666)	-1.058 (1.121)
Medicaid Share $\times$ ATC = H	1.144 (1.358)	0.446 (0.530)
Medicaid Share $\times$ ATC = J	-0.145 (0.165)	-0.078 (0.089)
Medicaid Share $\times$ ATC = L	-0.321 (2.789)	-0.244 (2.115)
Medicaid Share $\times$ ATC = M	0.022 (1.127)	0.017 (0.858)
Medicaid Share $\times$ ATC = N	0.034 (0.074)	0.047 (0.101)
Medicaid Share $\times$ ATC = R	0.298 (0.217)	0.178 (0.130)
Medicaid Share $\times$ ATC = S	8.052* (3.737)	6.473* (3.004)
Num. obs.	561 600	561 600
R2	0.235	0.235

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The FDA approval results contrast starkly from the patent pattern. Significant positive responses are observed for alimentary tract and metabolism, dermatologicals, and sensory organs, while no therapeutic class exhibits a significant negative response. The coefficient for sensory organs is large in magnitude but imprecisely estimated given the small number of approvals in that class.

Despite a strong negative patenting response in alimentary tract and metabolism classes, positive FDA approval coefficients are observed in these areas in response to the ACA. This divergence is consistent with the pipeline interpretation, where FDA approvals observed during the post-expansion period most likely reflect R&D investments made prior to the Medicaid expansion, so the upstream patent decline has not yet propagated to the approval stage. Additionally, dermatologicals have a positive approval response while having an insignificant patent coefficient, indicating that approval estimates do not cleanly map onto patent responses within therapeutic areas. Both patents and FDA approvals have positive coefficients for pharmaceuticals relating to sensory organs, but the large standard errors suggest that this pattern should be interpreted with caution.

Several therapeutic classes display negative point estimates, but none reach statistical significance. This reinforces the earlier finding that the upstream decline in innovation has not yet reached the end stage of approvals. Consequently, the therapeutic-level heterogeneity identified in patenting has not yet been reflected in regulatory approvals, consistent with the long development horizons separating early-stage innovation from drug approval.

4 PDL Counterfactual Analysis

	Patents	FDA Approvals
Post $\times$ P&T Pass (Clinical Only)	-11.399* (4.663)	1.002 (0.633)
Post $\times$ Price Pass (Price Only)	-7.386*** (2.173)	0.057 (0.881)
Post $\times$ In PDL (Both Stages)	15.549** (5.215)	-0.112 (1.174)
Num. obs.	5 281 696	151 200
R2	0.535	0.609

* p < 0.05, ** p < 0.01, *** p < 0.001

References

Garthwaite, C., Sachs, R., and Stern, A. D. (2021). Which markets (don't) drive pharmaceutical innovation? evidence from u.s. medicaid expansions. Working Paper 28755, National Bureau of Economic Research.

A Regression Tables

A.1 RQ1: Overall Innovation Response

	FEOLS		FEPOIS	
	Ex Ante	Ex Post	Ex Ante	Ex Post
2010 $\times$ Medicaid Share	−0.041 (0.038)	−0.033 (0.035)	0.120 (0.154)	0.129 (0.157)
2011 $\times$ Medicaid Share	−0.017 (0.036)	−0.010 (0.033)	0.218 (0.140)	0.240 (0.145)
2012 $\times$ Medicaid Share	0.002 (0.029)	0.004 (0.027)	0.115 (0.108)	0.124 (0.115)
2014 $\times$ Medicaid Share	−0.022 (0.027)	−0.018 (0.024)	−0.092 (0.108)	−0.086 (0.106)
2015 $\times$ Medicaid Share	−0.086** (0.030)	−0.088** (0.027)	−0.300* (0.118)	−0.367** (0.119)
2016 $\times$ Medicaid Share	−0.125*** (0.034)	−0.123*** (0.032)	−0.435*** (0.124)	−0.515*** (0.129)
2017 $\times$ Medicaid Share	−0.114** (0.036)	−0.122*** (0.034)	−0.397** (0.131)	−0.533*** (0.134)
2018 $\times$ Medicaid Share	−0.132*** (0.039)	−0.135*** (0.037)	−0.415** (0.149)	−0.556*** (0.157)
2019 $\times$ Medicaid Share	−0.189*** (0.041)	−0.194*** (0.039)	−0.784*** (0.154)	−0.998*** (0.167)
2020 $\times$ Medicaid Share	−0.220*** (0.043)	−0.225*** (0.041)	−0.998*** (0.173)	−1.293*** (0.190)
2021 $\times$ Medicaid Share	−0.220*** (0.043)	−0.222*** (0.040)	−0.989*** (0.170)	−1.313*** (0.188)
2022 $\times$ Medicaid Share	−0.233*** (0.043)	−0.233*** (0.040)	−1.137*** (0.186)	−1.564*** (0.203)
2023 $\times$ Medicaid Share	−0.243*** (0.044)	−0.238*** (0.040)	−1.324*** (0.201)	−1.803*** (0.224)
2024 $\times$ Medicaid Share	−0.252*** (0.046)	−0.250*** (0.042)	−1.474*** (0.261)	−2.021*** (0.266)
2025 $\times$ Medicaid Share	−0.228*** (0.044)	−0.212*** (0.041)	−1.110*** (0.229)	−1.692*** (0.262)
Num. obs.	5 281 696	5 281 696	1 300 944	1 300 944
R2	0.534	0.534		

* p < 0.05, ** p < 0.01, *** p < 0.001

	FEOLS		FEPOIS	
	Ex Ante	Ex Post	Ex Ante	Ex Post
2010 × Medicaid Share	0.215 (0.463)	0.286 (0.450)	0.237 (0.636)	0.453 (0.869)
2011 × Medicaid Share	−0.017 (0.381)	0.336 (0.413)	0.326 (0.446)	1.138 (0.687)
2012 × Medicaid Share	0.208 (0.412)	0.414 (0.455)	0.643 (0.615)	1.250 (0.863)
2014 × Medicaid Share	−0.321 (0.463)	−0.324 (0.533)	−0.545 (0.638)	−0.711 (0.996)
2015 × Medicaid Share	−0.053 (0.477)	0.005 (0.420)	−0.434 (0.632)	−0.431 (0.725)
2016 × Medicaid Share	0.414 (0.423)	0.550 (0.413)	−0.834 (0.478)	−0.788 (0.660)
2017 × Medicaid Share	1.421* (0.677)	1.635* (0.688)	−0.495 (0.480)	−0.278 (0.735)
2018 × Medicaid Share	0.999 (0.550)	1.240* (0.560)	−0.772 (0.530)	−0.595 (0.713)
2019 × Medicaid Share	0.554 (0.538)	0.400 (0.497)	−0.381 (0.565)	−0.625 (0.764)
2020 × Medicaid Share	0.732 (0.509)	0.652 (0.529)	−0.733 (0.574)	−0.916 (0.839)
2021 × Medicaid Share	0.137 (0.632)	0.218 (0.573)	−0.867 (0.729)	−0.909 (0.910)
2022 × Medicaid Share	0.365 (0.645)	0.418 (0.590)	−0.637 (0.594)	−0.657 (0.775)
2023 × Medicaid Share	0.846 (0.641)	0.842 (0.576)	−0.566 (0.511)	−0.618 (0.695)
2024 × Medicaid Share	0.258 (0.523)	0.137 (0.502)	−0.680 (0.573)	−0.969 (0.854)
2025 × Medicaid Share	−0.137 (0.510)	−0.380 (0.531)	−0.668 (0.580)	−1.254 (0.892)
Num. obs.	151 200	151 200	42 176	42 176
R2	0.609	0.609		

* p < 0.05, ** p < 0.01, *** p < 0.001

A.2 RQ2: Medicaid Demand and PDL Exposure

	FEOLS: Ex Ante		FEOLS: Ex Post	
	Medicaid Market Share	PDL Exposure	Medicaid Market Share	PDL Exposure
2010	−0.041 (0.038)	1.546 (2.018)	−0.033 (0.035)	1.546 (2.018)
2011	−0.017 (0.036)	0.734 (1.642)	−0.010 (0.033)	0.734 (1.642)
2012	0.002 (0.029)	1.079 (1.289)	0.005 (0.027)	1.079 (1.289)
2014	−0.022 (0.026)	4.161* (1.890)	−0.017 (0.024)	4.161* (1.890)
2015	−0.086** (0.030)	0.836 (1.480)	−0.088** (0.027)	0.836 (1.480)
2016	−0.126*** (0.034)	−1.554 (1.620)	−0.123*** (0.032)	−1.554 (1.620)
2017	−0.114** (0.036)	−1.342 (1.890)	−0.123*** (0.034)	−1.342 (1.890)
2018	−0.132*** (0.039)	−2.786 (1.644)	−0.136*** (0.037)	−2.787 (1.644)
2019	−0.189*** (0.041)	−3.806* (1.912)	−0.195*** (0.039)	−3.807* (1.912)
2020	−0.220*** (0.043)	−4.649* (1.894)	−0.227*** (0.041)	−4.650* (1.894)
2021	−0.220*** (0.043)	−5.266** (1.687)	−0.224*** (0.040)	−5.267** (1.687)
2022	−0.233*** (0.044)	−6.029** (1.941)	−0.235*** (0.041)	−6.029** (1.941)
2023	−0.243*** (0.044)	−6.925*** (1.851)	−0.240*** (0.041)	−6.926*** (1.851)
2024	−0.252*** (0.046)	−6.484*** (1.788)	−0.252*** (0.042)	−6.485*** (1.788)
2025	−0.227*** (0.044)	−4.588** (1.549)	−0.212*** (0.041)	−4.589** (1.549)
Num. obs.	5 281 696		5 281 696	
R2	0.535		0.535	

* p < 0.05, ** p < 0.01, *** p < 0.001

	FEOLS: Ex Ante		FEOLS: Ex Post	
	Medicaid Share	PDL Exposure	Medicaid Share	PDL Exposure
2010	0.214 (0.461)	0.049 (0.807)	0.285 (0.448)	0.046 (0.807)
2011	-0.010 (0.382)	-0.243 (0.712)	0.344 (0.412)	-0.257 (0.711)
2012	0.171 (0.406)	1.402 (0.911)	0.371 (0.447)	1.395 (0.912)
2014	-0.333 (0.465)	0.403 (0.647)	-0.338 (0.536)	0.398 (0.647)
2015	-0.088 (0.478)	1.163 (0.881)	-0.037 (0.422)	1.158 (0.881)
2016	0.334 (0.426)	2.645 (1.683)	0.456 (0.418)	2.642 (1.682)
2017	1.364* (0.680)	1.923 (0.994)	1.566* (0.690)	1.924 (0.992)
2018	0.949 (0.552)	1.684 (1.021)	1.181* (0.563)	1.683 (1.021)
2019	0.519 (0.539)	1.164* (0.592)	0.359 (0.498)	1.171* (0.591)
2020	0.687 (0.514)	1.579 (0.942)	0.598 (0.537)	1.585 (0.942)
2021	0.038 (0.640)	3.114* (1.394)	0.103 (0.583)	3.110* (1.393)
2022	0.333 (0.648)	0.994 (0.655)	0.381 (0.593)	0.994 (0.653)
2023	0.809 (0.649)	1.185 (1.738)	0.798 (0.583)	1.191 (1.737)
2024	0.272 (0.523)	-0.452 (0.590)	0.154 (0.501)	-0.447 (0.589)
2025	-0.109 (0.506)	-1.118 (0.908)	-0.347 (0.525)	-1.113 (0.908)
Num. obs.	151 200		151 200	
R2	0.609		0.609	

* p < 0.05, ** p < 0.01, *** p < 0.001

B Therapeutic-Class Substitutability

Table 1: Average annual distinct NDC count by ATC class

ATC	Therapeutic class	Mean annual n_c
N	Nervous system	7351
C	Cardiovascular system	4297
A	Alimentary tract and metabolism	3407
J	Anti-infectives for systemic use	2332
D	Dermatologicals	1771
R	Respiratory system	1672
M	Musculo-skeletal system	1407
B	Blood and blood-forming organs	1389
L	Antineoplastic and immunomodulating	932
G	Genito-urinary and sex hormones	906
H	Systemic hormonal preparations	768
S	Sensory organs	579
V	Various	211
P	Antiparasitic products	165